# Orthogonal Row-Sector Rebound Dynamics in Terminal Zeta-Zero Spacing Residuals

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**Abstract**

We identify a row-sector rebound structure in terminal adjacent-zero spacing residuals near a $\sim 10^3$-row sector boundary identified through an unbiased window-scale sweep, without prior specification of the active modulus. Under a base-6 row-sector decomposition, the structure is amplitude-dominant for modulus 1009, while modulus 990 remains co-significant under permutation guardrails. Follow-on audits reject simple $h_1$-corridor gating and do not confirm local $h_1\times$ sector modulation. A synthetic null calibration shows that apparent growth of interaction strength under finer local-$h_1$ binning matches bin-resolution sparsity inflation rather than hidden phase-conditioned structure. Thus the row-sector rebound and the previously published $h_1$ packet organization are empirically distinct on the observed terminal horizon.

**Keywords:** Riemann zeta zeros; zero spacing; row-sector rebound; tail-sign polarity; empirical audit; terminal band; observed-horizon result.

## 1 Introduction and Relation to Prior Work

The principal paper established that rare extreme compressions in adjacent Riemann zeta-zero spacing organize in the first logarithmic harmonic

$$h_1(T) = \left\{ \frac{\log(T/2\pi)}{\pi} \right\}, \tag{1}$$

stably across resolution ablation and with unusual tightening in the terminal observational band [1]. That result concerns the *phase location* of compression packets: where in logarithmic phase space rare compressions prefer to land. The broader statistical context includes the classical zero-counting and spacing background of the Riemann zeta function [2, 4, 5], and the computations build on tabulated zero ordinates such as those made available by Odlyzko [3].

The present paper addresses a different question. Within the terminal residual tail population, how are compression and expansion events distributed across row-index sectors? This question is structurally distinct from the $h_1$ concentration question and requires a separate instrument. Rather than asking where compression packets cluster in a continuous logarithmic coordinate, we ask how terminal tail polarity is distributed across a discrete row-index decomposition.

The main result is a row-sector rebound structure: compression and expansion tails are not distributed identically across row-index sectors. Among the tested decompositions, modulus 1009 is the most consistent sector-amplitude ruler, while modulus 990 remains co-significant under permutation guardrails. Five follow-on audits establish that this sector structure is empirically orthogonal to the previously published $h_1$ organization on the observed terminal horizon: it is not gated by $h_1$ corridor membership, not confirmed as a local $h_1\times$ sector interaction, and not explained by bin-resolution artifacts.

Table 1 summarizes the relationship between the published work and the present continuation.

Table 1: Structural distinction between the published paper and the present continuation.

| | **Published paper** | **Present paper** |
|---|---|---|
| Question | Where do rare extreme compressions cluster? | How is terminal tail polarity distributed across row-index sectors? |
| Coordinate | $h_1(T)$ logarithmic phase | Row-index sector modulo $M$, base-6 decomposition |
| Primary finding | $h_1$ phase concentration and narrow Arc80 corridor | Sector amplitude asymmetry, strongest under modulus 1009 |
| Terminal effect | Compression packet tightening in the terminal band | Permutation-significant sector polarity and rebound structure |
| Relationship | Not applicable | Empirically orthogonal to $h_1$ organization under tested coupling forms |
| Horizon | Published $h_1$ corridor horizon | Terminal board used in P74–P78, with explicit observed-horizon limitation |

This paper does not claim that $h_1$ controls or gates the row-sector rebound, that local $h_1\times$ sector modulation is confirmed, or that modulus 1009 is uniquely proven as a physical modulus. The scope of each positive finding is bounded by the explicit null testing reported in Sections 3 and 4. The negative findings are part of the result: they prevent the continuation structure from being folded back into the $h_1$ packet-corridor claim.

## 2 Data and Instrument

### 2.1 Source Board

All computations use the seam-stitched terminal checkpoint parquet from the principal paper,

`df_master_terminal_checkpoint_v16c_8000_31p5B.parquet`,

containing 935,111 rows with altitude range approximately 28,499.5M to 30,610.0M. Throughout this paper, all claims are empirical claims on this observed terminal horizon unless explicitly stated otherwise.

The suffix `31p5B` in the filename is a downloader target ceiling, not a claim that tabulated data extend to 31.5B. The closure audit P82 verified that the loaded parquet covers the complete available LMFDB terminal sequence for this campaign, with altitude range 28.499547B to 30.610043B, monotonic altitude ordering, no gap exceeding three times the mean local spacing, and SHA-256 checksum `c80b841b1ebd0a602b45db9fa0544499fc6781858b70bbb1ec0940aeffe0fa76`.

## 2.2 Residual Construction

For a window of size $W$ centered at row midpoint $r_{\text{mid}}$, the local spacing metric is computed from one of two residual-ratio coordinates:

$$\text{Gap\_to\_Pure\_Ratio} = \frac{\Delta_k}{s_{\text{pure}}(T_k)}, \tag{2}$$

$$\text{Gap\_to\_Fitted\_Ratio} = \frac{\Delta_k}{\widehat{s}(T_k)}, \tag{3}$$

where $s_{\text{pure}}$ is the pure logarithmic spacing spine and $\widehat{s}$ is the fitted spine used in the principal audit chain. For each window family, altitude drift is removed by selecting a best-BIC baseline from the audit model family, and residuals are converted to robust $z$-scores.

The primary tail threshold is

$$|z| \geq 2.575, \tag{4}$$

with a sensitivity threshold

$$|z| \geq 2.326. \tag{5}$$

Compression tails satisfy $z \leq -2.575$ and expansion tails satisfy $z \geq +2.575$ in the primary analysis.

## 2.3 Row-Sector Coordinate

For modulus $M$ and window row midpoint $r_{\text{mid}}$, define the base-6 row sector

$$s_{6,M}(r_{\text{mid}}) = \left\lfloor 6 \cdot \frac{r_{\text{mid}} \bmod M}{M} \right\rfloor \in \{0, 1, 2, 3, 4, 5\}. \tag{6}$$

The tested moduli are

$$M \in \{990, 997, 1009, 1080\}. \tag{7}$$

The primary stride is `stride_mod6`; the sensitivity stride is `stride_W_4`. The window range used for the final calibration chain is

$$W \in \{900, 925, 930, 950, 975, 1000, 1009, 1050, 1100, 1125, 1200, 1250, 1300\}. \tag{8}$$

## 2.4 Frozen Parameter Registry

The frozen registry records the altitude split point 30,000M, primary and sensitivity tail thresholds, moduli tested, stride modes, local $h_1$ bin counts, P78 cell-count floors, and the literal-claim matrix rule. In particular, P78 uses a minimum mean cell-count floor of 15 and a minimum cell count of 5 when interpreting interaction ratios. These parameters are recorded in the patched frozen parameter registry and P79B manifest (timestamp `79btimestamp`) before final paper-language assignment.

# 3 Audit Chain

The empirical case is built through five sequential audits, P74–P78, followed by the P79/P79B claim freeze. Each audit uses a pre-specified read rule and a separate null or calibration structure. Figure 1 presents the final literal-claim evidence matrix. Each cell answers whether the audit supports the claim as written, rather than whether the audit supports the final ledger disposition.

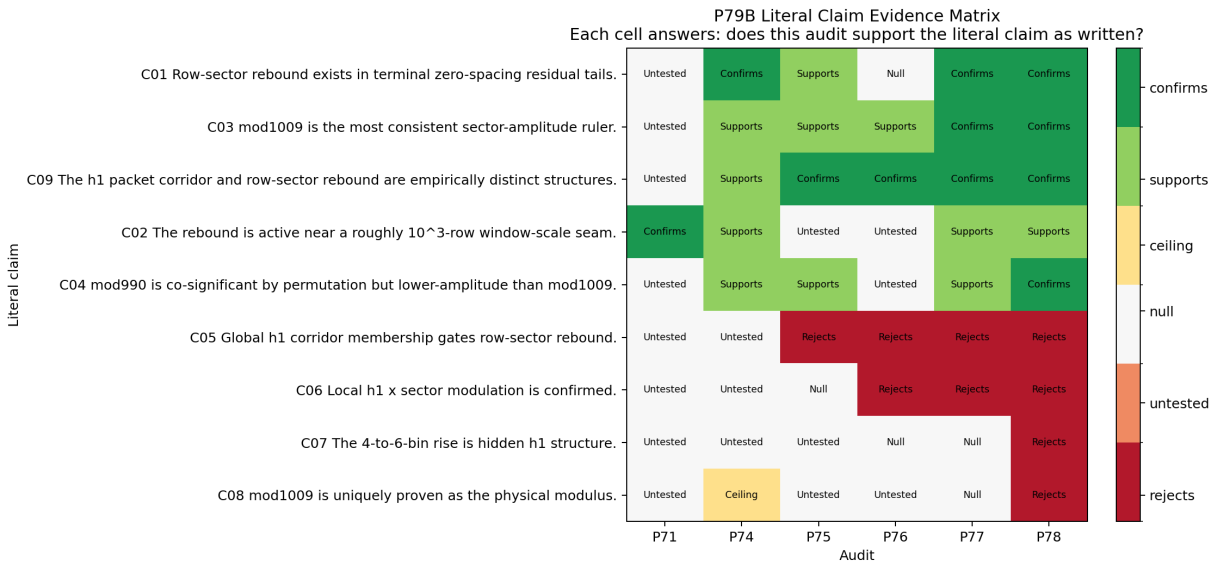


Figure 1: P79B literal claim evidence matrix. Each cell answers: does this audit support the literal claim as written? For rejected claims, an audit that contradicts the claim is coded as *rejects*, not as support for the rejection.

### 3.1 P74: Unbiased Window-Scale Rebound Sweep

P74 conducted an unbiased sweep over window sizes and sector configurations to identify whether a rebound seam exists without pre-specifying the active modulus. The moduli 990, 997, 1009, and 1080 were identified as the strongest candidates from the P74 sweep result and then carried forward without reselection. The strongest permutation-tested cluster occurred near the $\sim 10^3$-row scale, with top configurations concentrated in the $W \approx 925$–$1300$ band under base-6 sectorization. The audit verdict was

`BALANCE_SEAM_REBOUND_PEAK_SURVIVES_SWEEP`.

This established the rebound seam as a candidate structure, but did not by itself determine whether the structure shares a chassis with the previously published $h_1$ organization.

### 3.2 P75: $h_1$-Gated Rebound Audit

P75 tested whether the row-sector rebound is stronger inside the published $h_1$ corridor than in shoulder regions. Let

$$\Delta_{h_1} = \text{rebound\_score}_{\text{core}} - \text{rebound\_score}_{\text{shoulder}}. \tag{9}$$

The corridor was loaded from the published manifest without retuning. Four nulls were used: sector shuffle within $h_1$ bins, $h_1$-bin shuffle within sector and altitude bins, circular row-phase shift, and altitude-matched $h_1$ assignment. Across the tested configurations, $\Delta_{h_1}$ did not survive the null battery. The audit verdict was

`ORTHOGONAL_ROW_REBOUND_CONFIRMED`.

Thus, the row-sector rebound is not supported as a simple function of membership in the published $h_1$ corridor.

### 3.3 P76: Global $h_1\times$ Sector Decomposition

P76 constructed a two-dimensional imbalance map over global $h_1$ bins and base-6 row sectors. For $h$-bin $a$ and sector $s$, define the tail-sign imbalance

$$I(a,s) = \log\left(\frac{E(a,s)+1/2}{C(a,s)+1/2}\right), \tag{10}$$

where $E(a,s)$ and $C(a,s)$ are expansion and compression counts in the cell. The interaction residual is

$$R(a,s) = I(a,s) - \overline{I}(a,\cdot) - \overline{I}(\cdot,s) + \overline{I}. \tag{11}$$

Here $\overline{I}(a,\cdot)$ denotes the mean of $I(a,s)$ over $s$, $\overline{I}(\cdot,s)$ denotes the mean over $a$, and $\overline{I}$ denotes the grand mean. The strength ratio $S_{\text{int}}/S_{\text{main}}$ was used to compare interaction residual magnitude to the main-effect scale.

The important diagnostic from P76 was structural: the terminal tail population was so tightly concentrated in global $h_1$ that most global $h_1$ bins were empty. Consequently, the global interaction map had no vertical leverage, and $S_{\text{int}}/S_{\text{main}}$ collapsed to numerical zero. The audit verdict was

`SECTOR_DOMINANT_H1_WEAK_MODULATOR`.

P76 therefore motivated a local-$h_1$ instrument rather than a global-bin instrument.

### 3.4 P77: Local-$h_1$ Leverage and Sector Engine Audit

P77 replaced global $h_1$ bins with local quantile bins computed within the terminal tail support. The primary local instrument used four quantile bins; the sensitivity instrument used six. This restored vertical leverage while preserving the sector coordinate from Equation (6).

The sector engine survived tail-label shuffle for the leading moduli, especially moduli 1009 and 990. The local $h_1\times$ sector interaction residual did not survive null testing. The mod1009 consistency check ranked modulus 1009 first by $S_{\text{sector}}$ amplitude in the tested primary lanes. The audit verdict was

`MOD1009_SECTOR_ENGINE_CONFIRMED_H1_ORTHOGONAL`.

The P77 local imbalance maps are useful descriptive objects, but they are not promoted to confirmed local-$h_1$ interaction claims because their interaction component did not survive the null battery.

### 3.5 P78: Bin-Resolution Null Calibration

P78 tested the apparent rise in $S_{\text{int}}/S_{\text{sector}}$ under finer local-$h_1$ binning. The bin count was swept over

$$B \in \{2,3,4,5,6,8,10,12\}, \tag{12}$$

with matched tail-label nulls and a synthetic Bernoulli-null control on the real geometry. The key observation was that the synthetic null produces the same rising interaction-ratio shape as bin count increases. Thus the 4-to-6-bin rise is a sparsity and bin-resolution effect, not a hidden local-$h_1$ signal.

The P78 primary calibration curve is shown in Figure 2. Observed curves remain within the null envelope, and polynomial tests do not show significant excess intercept or curvature for the primary mod1009 lane. The audit verdict was

`BIN_RESOLUTION_INFLATION_CONFIRMED`,

with the secondary finding

MOD1009_AMPLITUDE_LEADER_CONFIRMED.

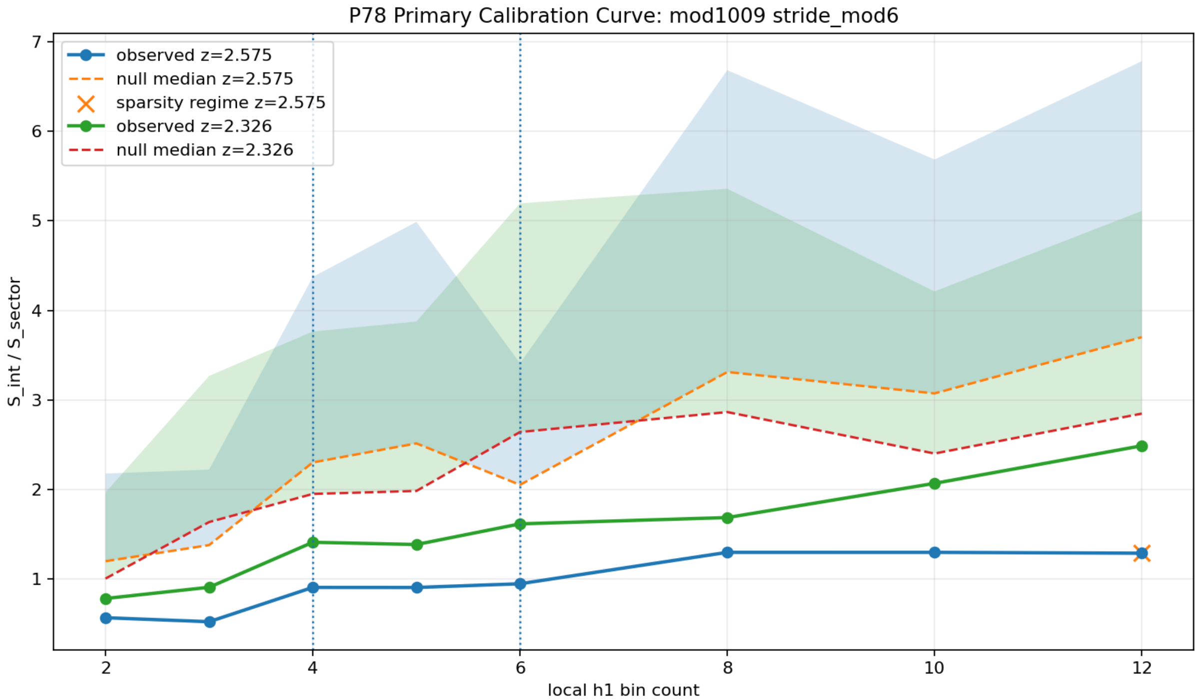


Figure 2: P78 primary calibration curve for the mod1009, `stride_mod6` lane. The observed local-$h_1$ interaction-ratio curve remains within the matched null envelope, indicating bin-resolution inflation rather than confirmed local-$h_1$ modulation.

# 4 Results Summary

The final claim hierarchy is generated by the P79B hardened ledger. It separates publishable claims, supplemental claims, and rejected or quarantined claims.

## 4.1 Level 1: Confirmed Claims

**C01. Row-sector rebound structure exists.** Terminal adjacent-zero spacing residuals exhibit a row-sector rebound structure: compression and expansion tails are not distributed identically across row-index sectors. This claim is supported by the unbiased rebound sweep in P74, the sector-engine null tests in P77, and the P78 amplitude guardrail. Empirical on the observed terminal horizon only; extension to greater heights is not claimed.

**C03. Modulus 1009 is the most consistent sector-amplitude ruler.** Among the tested row-sector decompositions, modulus 1009 is the most consistent sector-amplitude ruler across 32 of 32 tested configurations spanning two metrics, two stride modes, and eight local bin counts, while modulus 990 remains co-significant by permutation guardrail. This does not prove modulus 1009 is the unique physical modulus; p-value rankings are ceiling-bounded and modulus 990 remains co-significant.

**C09. The $h_1$ packet corridor and row-sector rebound are empirically distinct structures.** The previously published $h_1$ packet organization and the row-sector rebound are empirically distinct on the observed terminal horizon. The rebound is not gated by corridor membership, not confirmed as a local $h_1\times$ sector modulation, and not explained by bin-resolution artifacts. Distinct on the observed terminal horizon does not preclude a deeper common mechanism at greater heights or under a different mathematical framework. The current evidence rules out the tested coupling forms; it does not rule out untested ones.

### 4.2 Level 2: Supported Claims

**C02. The rebound is active near a roughly $10^3$-row seam.** The rebound structure is most naturally studied near a roughly $10^3$-row sector boundary identified through an unbiased window-scale sweep, without prior specification of the active modulus. This is an observed-horizon scale, not a universal correlation length theorem.

**C04. Modulus 990 remains co-significant but lower-amplitude than modulus 1009.** Modulus 990 remains co-significant under permutation guardrails, but modulus 1009 carries larger sector amplitude on the observed horizon. P-value ceiling effects prevent unique ranking by significance alone.

### 4.3 Level 4: Rejected or Quarantined Claims

The following claims were explicitly tested and rejected, or quarantined as too strong for the available evidence:

- **C05.** Global $h_1$ corridor membership gates row-sector rebound. Rejected by P75 and contradicted by subsequent interaction audits.
- **C06.** Local $h_1\times$ sector modulation is confirmed. Rejected or unconfirmed by P76–P78.
- **C07.** The 4-to-6-bin rise is hidden $h_1$ structure. Rejected by the P78 synthetic and matched null calibration.
- **C08.** Modulus 1009 is uniquely proven as the physical modulus. Quarantined: amplitude leadership is confirmed; uniqueness is not.
- **C11.** A large-window mod990/mod1009 sector-amplitude crossover indicates scale-separated sector engines. Rejected by P81. The apparent P80 crossover was not reproduced under the pre-specified P81 criterion and is attributed to window-size-dependent sparsity variation rather than modulus-specific scale separation.

## 5 Discussion

The audit chain resolves a tension raised by the principal paper. Does the terminal band's strong $h_1$ phase concentration reflect a single organizing mechanism that also governs compression/expansion polarity, or are there two distinct structures sharing the same terminal real estate?

The evidence favors the second interpretation. The $h_1$ organization and the row-sector rebound coexist in the terminal band, but they do not appear to share the tested coupling forms. The $h_1$ coordinate organizes packet location in a continuous logarithmic phase. The row-sector instrument detects a discrete tail-sign polarity structure across row-index sectors. These are orthogonal empirical structures on the observed terminal horizon.

This negative result is structurally informative. It prevents the sector rebound from being misdescribed as merely a shadow of the $h_1$ corridor. It also prevents the amplitude leadership of modulus 1009 from being promoted into a uniqueness claim. The resulting model is narrower but stronger: a terminal row-sector rebound exists; modulus 1009 is the most consistent amplitude ruler; modulus 990 remains co-significant; and the tested $h_1$ coupling forms fail.

### 5.1 What Would Change These Conclusions

Disambiguation of modulus 1009 and modulus 990 by p-value would require a permutation design with sufficient non-ceiling separation, extended altitude data beyond the current horizon, or a larger tail population under altitude controls. None of these are available on the current horizon. Here "extended altitude data" means newly generated source data beyond the current LMFDB terminal ordinate of approximately 30.610B, not discovery of an existing hidden 31.5B parquet. The filename suffix `31p5B` was a retrieval ceiling only; P82 confirmed the loaded parquet is the complete available sequence for this campaign.

The local $h_1\times$ sector interaction question is also bounded by the available tail population. P78 rules out the tested binning families and tail thresholds, not every conceivable higher-order relation. A larger tail population could provide enough cell-level resolution to test weaker modulation forms that the current instrument cannot detect.

### 5.2 Observed-Horizon Scope

All continuation claims are confined to the observed terminal band. The row-sector structure and the $h_1$ orthogonality result are empirical laws on this observed horizon. No asymptotic extension is claimed.

### 5.3 Post-Freeze Closure Audits

After the P79B claim freeze, two narrow post-freeze audits were run to test possible interpretive hazards. P80 tested a simplified row-window residual proxy against an altitude-matched GUE null; the verdict `GUE_ARCHITECTURE_MARGINAL` applies to that proxy instrument only and does not rerun or refute the frozen packet extractor of the principal $h_1$ paper. P81 tested a possible large-window mod990/mod1009 sector-amplitude crossover suggested by the P80 sensitivity sweep. Under a denser window ladder, altitude stratification, threshold sweep, null-sigma separation, and nearby-modulus controls, the crossover was not reproduced (`CROSSOVER_NOT_REPRODUCED`). P82 then reconciled the terminal horizon, verified the parquet provenance, and closed the branch. These post-freeze audits do not alter the publishable claims C01, C03, or C09.

## 6 Limitations

The primary claims are empirical and confined to the observed horizon. The sector-amplitude leadership of modulus 1009 and co-significance of modulus 990 are properties of the terminal band tested here; behavior at greater heights would require fresh data.

The permutation ceiling affects p-value interpretation for the strongest sector configurations. P-value ranking between modulus 990 and modulus 1009 cannot be used to crown a unique winner on the current evidence.

The local $h_1$ interaction tests operate on sparse cells. With the current tail population divided across base-6 sectors and local $h_1$ bins, weak interactions may be undetectable. P78 rules out strong bin-resolution interaction in the tested families; it does not rule out weak local phase conditioning below the current instrument resolution.

The $\sim 10^3$-row sector seam was identified empirically through the P74 window-scale sweep. The tested moduli are reported as part of the audit protocol, and claims are phrased accordingly: modulus 1009 is an amplitude leader among tested decompositions, not a uniquely proven physical modulus.

## 7 Conclusion

Terminal adjacent-zero spacing residuals exhibit a row-sector rebound structure, amplitude-dominant under modulus 1009, co-significant under modulus 990, and empirically orthogonal to the previously published $h_1$ packet organization on the observed terminal horizon. Five sequential audits establish this structure and bound what it is not: the rebound is not gated by $h_1$ corridor membership, local $h_1\times$ sector modulation is not confirmed, and the apparent growth of interaction strength under finer binning is explained by sparsity inflation. Two post-freeze audits further bound the claim: P80 is retained as a simplified proxy caution, and P81 does not reproduce the tested large-window mod990/mod1009 crossover on the current LMFDB horizon. The continuation claim is deliberately narrow: two empirical structures coexist in the terminal band on the observed horizon, and the tested coupling forms between them do not survive null controls.

## Data and Artifact Availability

All audit outputs, manifests, the frozen parameter registry, the literal claim evidence matrix, and the hardened safe-language document are archived in the associated artifact bundle under `ForgeV16c_P79B_LEDGER_HYGIENE_outputs`, with timestamps as recorded in the P79B manifest . The master parquet and seam-stitched board are as described in the principal paper [1].

Post-freeze closure artifacts are archived in:
`ForgeV16c_P82_CLOSURE_outputs`, including `p82_final_closure_82timestamp_manifest.json`, the terminal parquet provenance record with SHA-256 checksum, frozen-claims registry, and audit provenance chain. P81 crossover outputs are archived in `ForgeV16c_P81_CROSSOVER_outputs` with verdict `CROSSOVER_NOT_REPRODUCED`.

## Use of AI Tools

Generative AI tools were used for audit design review, code critique, adversarial claim evaluation, and draft assistance. All research design decisions, audit execution, interpretation, and final manuscript responsibility remain with Jeffery Huckstead.

## Author Contributions

Jeffery Huckstead conceived and executed the audit chain, designed the claim ledger and freeze protocol, and wrote the manuscript.

## Acknowledgments

The author thanks the structured audit-review and adversarial critique process that shaped the ForgeV16c analytical chain.

## References

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